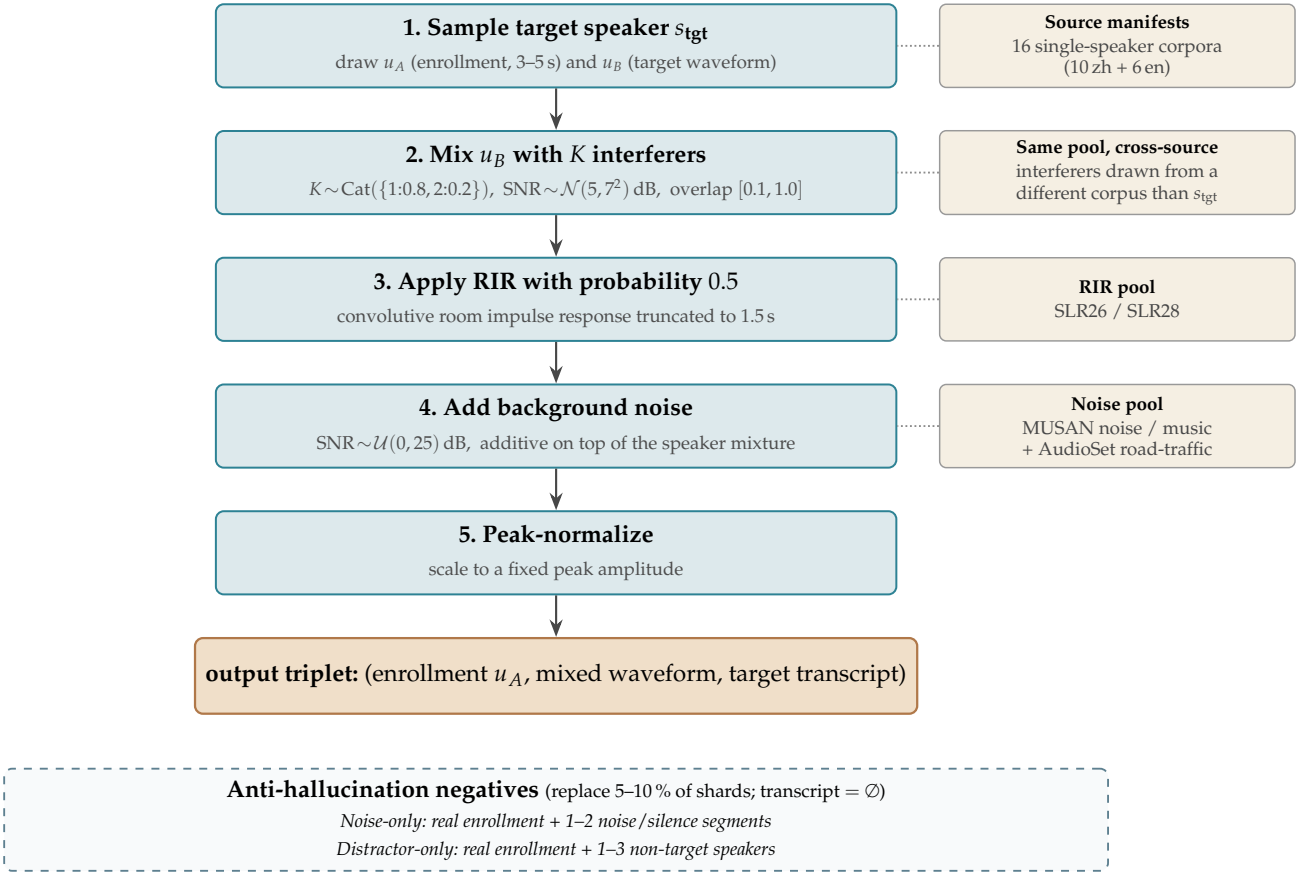

1. Sample target speaker s_{tgt}
draw u_A (enrollment, 3–5 s) and u_B (target waveform)
Source manifests
16 single-speaker corpora
(10 zh + 6 en)
2. Mix u_B with K interferers
$K \sim \mathrm{Cat}(\{1{:}0.8, 2{:}0.2\})$, $\mathrm{SNR} \sim \mathcal{N}(5, 7^2)$ dB, overlap $[0.1, 1.0]$
Same pool, cross-source
interferers drawn from a
different corpus than s_{tgt}
3. Apply RIR with probability 0.5
convolutive room impulse response truncated to 1.5 s
RIR pool
SLR26 / SLR28
4. Add background noise
$\mathrm{SNR} \sim \mathcal{U}(0, 25)$ dB, additive on top of the speaker mixture
Noise pool
MUSAN noise / music
+ AudioSet road-traffic
5. Peak-normalize
scale to a fixed peak amplitude
output triplet: (enrollment u_A, mixed waveform, target transcript)
Anti-hallucination negatives (replace 5–10 % of shards; transcript = ∅)
Noise-only: real enrollment + 1–2 noise/silence segments
Distractor-only: real enrollment + 1–3 non-target speakers